***** WRITE YOUR NAME AND COHORT BELOW *****

UNIVERSITY OF CALIFORNIA BERKELEY
Haas School of Business

# MBA 201A—Economic Analysis for Business Decisions—Fall 2019

Professor Steve Tadelis

## MIDTERM EXAM

**Instructions:** The number in brackets (*e.g.,* [5]) indicates the points for each question. Total: 85 points. Note that you have 90 minutes to do the exam, so if you spend no more than 1 minute per point, you will have some time to check your work.

***Please Write Legibly.*** Briefly explain your answers (that is, don't just write "yes" or "no" and don't just write down a numerical answer without showing how you derived it). Write only on this exam and not on other sheets of paper. The last page is a blank sheet for your use.

****** Please sign the honor code oath at the bottom of this page. ******

**********

Honor Code:

I understand that this exam is an individual exercise. I swear on my honor that I have not consulted with another person or made use of notes or other materials on- or off-line during the exam.

Name:_____

Cohort:_____

**Short answer questions 1-3** [15 points]

These 3 questions require **short answers** (up to 5 lines/sentences). Use any graphs or calculations that will help your explanation and label graphs clearly.

1. [5] A San Francisco real estate developer wishes to build a new development but local government is generally opposed to new construction. It demands an environmental impact study that will yield an "impact score," which is an index number based on the impact the development will likely have on traffic, air quality, sewage and water usage, etc. "I'm convinced," says the developer, who has lots of experience, "that we'll get an impact score somewhere between 40 and 70." Given the local government's general hostility to new development, you know that there's a 35% chance that it will okay the development if the impact score is less than 50, a 5% chance that it will okay the development if the score is between 50 and 55, and no change that it will okay the development if the score is greater than 55. Your initial assessment indicates that any score between 40 and 70 is as likely as any other score between 40 and 70. The value of the development to the developer is $20,000,000. What is the maximum cost of the impact study such that it is still worthwhile for the developer to have one conducted? (The developer is risk neutral.)

2. [5] You are flying abroad when the person next to you in business class strikes up a conversation. Learning that you're a business student, he asks your advice about the following issue. "I'm trying to expand how much my company sells," he says. "But I'm having a deuce of a time—I cut price by 5% and increased volume by only 1%. I don't know, do you think I should be more aggressive on the price cutting; maybe going down another 5%?" What do you suggest to this person?

3. [5] B&B Shampoo buys a special seaweed extract for one of its shampoos from Joe's Seaweed in Rockland, Maine. Joe's makes the seaweed extract specially for B&B, and there are no other known consumers of the product. B&B signed a long-term contract with Joe's and has agreed to buy 1000 gallons of the extract a month for the next five years at a price of $200 per gallon. B&B overestimated how much seaweed extract it would need, so it currently has a surplus of approximately 300 gallons a month. In part to use the surplus extract, it has decided to launch a styling gel that will use 100 gallons of the extract. What are B&B's economic costs of using the seaweed extract to produce the new styling gel? Explain.

## *MULTI-PART QUESTIONS*

Question 4 [20 points]

A gas station in a more rural part of the state faces essentially no competition. Its weekly demand is given by 82,500 − 30,000$p$, where $p$ is the price per gallon that it charges. Currently, the gas station is charging $2.63 per gallon. Assume that is the profit-maximizing price. For all extents and purposes, the gas station's marginal cost of selling a gallon of gasoline is the wholesale price it pays per gallon. Suppose, due to recent events, the wholesale price of gasoline rises by 5%.

a. [10] What is the new profit-maximizing price for the gas station to charge its customers?

b. [5] By how much does the gas station's profit per week rise or fall?

c. [5] What does your analysis suggest about the validity of allegations that gas stations get rich when the price of oil rises?

Question 5 [20 points]

Chevron is facing a tough decision. A new oil project called "Haiti" is scheduled to produce its first commercial oil in mid-2020, yet it is still unclear how productive it will be. "Haiti is one of Chevron's five big projects," told Peter Robertson, vice chairman of the company's board to the Wall Street Journal. Still, it was unclear whether the project will result in the blockbuster success Chevron is hoping for. As for now, $2-billion were invested in the high-tech deep-sea platform, which suffices to perform early tests. Aside from offering information on the type of reservoir, the tests will produce enough oil to just cover the costs of the tests. Current estimates are that the price of oil will increase annually by 6%, which is just about the costs of capital for Chevron. Current oil prices are at about $70 a barrel. Following the test wells, Chevron

predicts one of three possible 5-year scenarios. The optimistic one is that Tahiti sits on one giant, easily accessible oil reservoir, in which case the company expects to extract 200,000 barrels a day after expending another $2.5-billion in platform set-up costs, with a cost of extraction at about $10 a barrel. A less rosy scenario, that is twice as likely as the optimistic one, is that Chevron would have to drill several more wells at an additional $500-million (above and beyond the $2.5-billion set-up costs), in which case production will be at around 100,000 barrels a day with a cost of extraction at about $30 a barrel. The worst-case scenario involves the oil tucked away in numerous pockets, requiring expensive water injection techniques which would include up-front costs of another $2-billion (above and beyond the $2.5-billion set-up costs), extraction costs of $50 a barrel, and production is estimated to be at about 60,000 barrels a day. Bill Varnado, Haiti's project manager, was quoted giving this least desirable outcome odds of 50%. It is safe to assume that after 5 years the wells will most likely produce at a level that will just cover costs.

a. [12] If the test-wells would not produce information about which of three possible scenarios will happen, should Chevron invest the set-up costs of $2.5- billion to be prepared to produce at whatever scenario is realized?

b. [8] If the test-wells do produce accurate information about which of three possible scenarios is true, what is the added value of performing these tests?

Question 6 [25 points]

At the 2009 Frankfurt Motor Show, the founder and CEO of Better Place, Shai Agassi, announced a newly expanded agreement with Renault that commits both companies to a volume of at least 100,000 electric cars in Israel and Denmark by 2016. In May 2011, after his commencement speech at the Berkeley-Haas MBA graduation event, Agassi offered you the job of running the franchise to distribute the needed lithium-ion batteries in Denmark for the next five years (January 2012 through December 2016).

You are now in the market to lease a warehouse outside of Copenhagen.

A small warehouse costs $1 million per year and a large warehouse costs $2 million per year. Current contracts show that your sales will be low in the first two years and that you will only need a small warehouse for these years. Reliable estimates show that there is a 0.3 probability that your sales will take off and that you will have high demand, in which case you will need a large space in years 3 through 5 (and a 0.7 probability that demand stays low and you will only need a small space in all years). If your sales take off and you face high demand, you could increase sales in year 3, but only if you have leased the large warehouse. Data from Better Place includes the following table (which applies to all years):

| | | **Total Annual Revenues** | **Annual Variable Costs** |
|---|---|---|---|
| **Small Warehouse** | Low Demand | 3,000,000 | 1,200,000 |
| | High Demand | 3,000,000 | 1,200,000 |
| **Large Warehouse** | Low Demand | 3,000,000 | 1,200,000 |
| | High Demand | 8,000,000 | 3,200,000 |

**NOTE:** Variable Costs include all costs *except* the warehouse lease cost. Profit is equal to Total Revenue – Variable Costs – Costs of the Lease. ***Assume that the interest rate is 0%, so no discounting applies.***

a. [10] Draw a decision tree to help you decide whether to lease a large of small warehouse. (Note that you cannot change warehouses, so whatever warehouse you choose you will have in all five years.)

b. [5] Assuming you are a risk-neutral expected-profit maximizing decision maker, should you lease a large warehouse or a small warehouse?

c. [5] Might your answer to part a) change if you were risk averse?

d. [10] A real estate company offers you an "exit-option lease." This is a lease for a large warehouse in the first two years, but it would allow, though not require you to downsize your operation to a small warehouse (less space in the original one) at the end of year 2, after you had learned whether demand would be high or low in years 3 through 5. Each year in the large warehouse would still have an annual cost of $2 million and each year in a small warehouse would still have an annual cost of $1 million. The exit-option lease, however, has two twists. First, it includes a signing fee of $50,000 upfront above and beyond the annual fees. Second, it includes an exercise fee of $500,000, which must be paid at the end of year 2 only if you decide to downsize the warehouse. Assume again that you are risk neutral. Should you sign this exit-option lease instead of the one you chose in part b)?